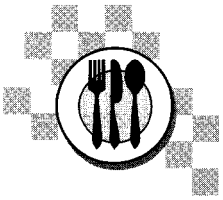
CHAPTER 4



Personal Hygiene and Food Handling

About This Chapter

In this chapter you will learn:

1. What personal hygiene is and why it is important
2. About parts of the body that need to be kept clean or covered
3. When and how to wash hands to keep food safe
4. How diseases are spread
5. How employees and employers can take responsibility for personal hygiene

Introduction

Food handlers can carry bacteria that can cause illness in people who eat the food handlers prepare. In fact, people are the most common source of food contamination. Hands, breath, hair, sweat, coughs, and sneezes all carry microorganisms. Even if a food handler does not feel sick, he or she could still be carrying microorganisms that can cause illness if they get into food.

The food industry spends a lot of time educating and training employees. Supervisors and workers need to understand the importance of protecting food. Foodborne disease outbreaks are very bad for business. A foodborne illness outbreak can cost about $75,000 per foodservice establishment. Investigation, cleanup, restaffing and restocking, wasted food products, settlements, and regulatory sanctions are all expensive. In multiunit restaurant chains, negative public opinion can affect sales even at units that were not involved in the outbreak.

PERSONAL HYGIENE

The word *hygiene* means using sanitary principles to maintain health. Personal hygiene refers to the cleanliness of a person's body and clothes. Food workers need to be healthy and clean to prepare safe food.

Employee Hygiene

Employees who are ill should not come to work. They should not touch food or equipment and utensils used to process, prepare, and serve food. Food can carry several illnesses, including:

- Respiratory diseases, e.g., colds, sore throats, pneumonia, scarlet fever, and tuberculosis
- Gastrointestinal diseases, e.g., vomiting, diarrhea, dysentery
- Typhoid fever
- Infectious hepatitis

After people recover from the disease, they often become carriers. This means that they still carry the disease-causing microorganisms in or on their body.

When employees are ill, they carry many more microorganisms, so they are much more likely to contaminate food. Symptoms of illnesses show that the number of microorganisms in the body is increasing, although sometimes they increase before symptoms appear. Anyone with a sinus infection, sore throat, nagging cough, or other cold symptom is probably carrying a heavy load of a virus. People who have diarrhea or an upset stomach are also probably carrying large numbers of microbes. Even after the symptoms have gone away, some of the microorganisms that caused the illness may stay in the person's body and could contaminate food. For example, an employee may carry *Salmonellae* for several months after recovering from salmonellosis. The virus that causes hepatitis may still be in the intestinal tract over 5 years after the symptoms are over. To understand why employees need good personal hygiene, it helps to look at different parts of the body that can be sources of microbial contamination. Figure 4.1 shows that bacteria live on different parts of the body and how handwashing affects the number of bacteria.

Skin

The skin constantly deposits sweat, oil, and dead cells on its outer surface. When these materials mix with dust, dirt, and grease, they form an ideal place for bacteria to grow. Therefore, bacteria from skin can contaminate food. If secretions build up and bacteria continue to grow, the skin can become itchy or irritated. Food handlers may rub or scratch the area and then transfer bacteria to food when they touch it. Contaminated food has a shorter shelf life or may cause foodborne illness. Washing

HANDS

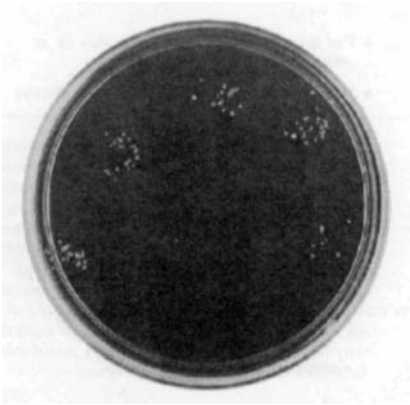
Let's see how handwashing affects the number of bacteria present



Bacteria grow on gelatin-like food (agar) in covered, sterile, plastic plates (petri dishes).



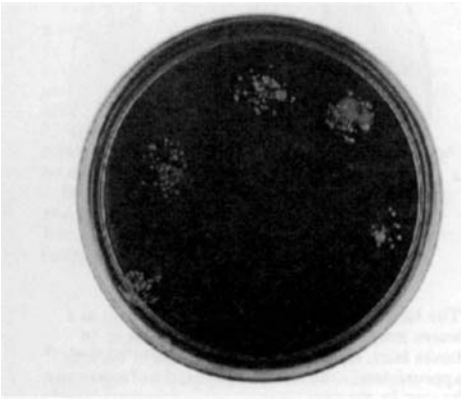
An unwashed hand that looks clean is touched to the agar.



After washing hands for 15 seconds with hot water and soap, bacteria are reduced in number.



The bacteria grow rapidly when kept at a warm temperature (98.6°F), 24 to 48 hours later, small colonies or clumps of bacteria approximately the size of a pinhead or larger can be seen in the agar.



The plate is incubated at 98.6°F for 24 hours. The heavy growth of white colonies indicates that this hand was not very clean and that millions of bacteria were present.



Washing hands with soap and water for another 15 seconds reduces the bacteria even more.

This picture shows the bacteria present in fingernail scrapings.

This plate shows what happens when the lips and nose are pressed against agar in a petri dish.

Hair is also a source of bacteria and has no place in food. It is unappetizing, unappealing, and adds bacteria to food. This picture shows the bacteria associated with human hair.

Each sneeze contains between 10,000 and 100,000 bacteria, and they are moved through the air at more than 200 mph.

FIGURE 4.1. Bacteria on your body and the effects of handwashing (from Cornell Cooperative Extension Publication: *Safe Food Preparation: It's In Your Hands*, Ithaca, N.Y., 1995.)

hands and bathing often reduces the number of microorganisms and dead cells that can be shed into the food.

Some types of bacteria do not grow on the skin, because the skin acts as a physical barrier and makes and releases chemicals that can destroy some microorganisms. This self-disinfecting system works best when the skin is clean.

The outer layer of skin (epidermis) contains cracks, crevices, and hollows where microorganisms like to live. Bacteria also grow in hair follicles and sweat glands. Because people use their hands a lot, they are often cut or callused or come in con-

tact with a wide variety of microorganisms. Bacteria that live on the skin are not easily removed. They live in the pores of the skin and are protected by oily secretions from the glands.

Poor skin care and skin disorders do more than affect appearance. They also cause bacterial infections like boils and impetigo.

Boils are severe local infections caused by infections in hair follicles or skin glands after the outer layer of skin (epidermis) is damaged, for example by irritating clothing. Staphylococci or other microorganisms multiply in the hair follicle or skin gland and produce a toxin that kills the cells around it and causes swelling and soreness. The body collects lymph, blood cells, and tissue cells in the infected area to counteract the toxin. The body forms a barrier around the boil to prevent the infection from spreading. A boil should never be squeezed. If it is squeezed, the infection may spread and cause a cluster of boils called a *carbuncle.* If staphylococci get into the bloodstream, they may be carried to other parts of the body, causing meningitis (infection of the membranes around the brain), bone infections, or other problems. Employees with boils should be very careful if they have to handle food, because the boil is a major source of pathogenic staphylococci. Employees should use a hand dip for disinfection after touching a boil or pimple. Boils can be prevented by keeping skin clean and wearing clean clothing.

Impetigo is a very infectious skin disease caused by *Staphylococci* bacteria. The infection spreads from one part of the body to another by touching. It is common in young people who do not wash often. Keeping the skin clean helps to prevent impetigo.

Fingers

Bacteria may be picked up by the hands when they touch dirty equipment, contaminated food, clothing, or parts of the body. Employees should wash hands frequently and use a hand-dip sanitizer after touching these things so that they do not contaminate food. Plastic gloves can be used, although food can be quickly contaminated if good handwashing and other hygienic practices are not used. Plastic gloves help prevent pathogenic bacteria on the fingers and hands from reaching the food. They also help those who watch the food being prepared to feel safe while eating it.

Fingernails

One of the easiest ways to spread bacteria is through dirt under the fingernails. Employees should never handle food if their fingernails are dirty. Food handlers should not have long fingernails or artificial fingernails. Washing the hands with soap and water removes transient bacteria (bacteria that have been picked up but are not growing and multiplying), and using a hand soap that contains an antiseptic or sanitizer controls resident bacteria (bacteria that live and grow there). A humectant (moisturizer) that contains alcohol can help to control and remove both transient and resident bacteria without irritating the hands.

Jewelry

Employees should not wear jewelry in food-processing or foodservice areas. It can get caught in machinery, causing a safety hazard. Also, contaminated jewelry can fall into or come in contact with food.

Hair

Hair carries microorganisms, especially staphylococci. Employees should wear a hairnet or hat while handling food and use a hand dip after scratching their heads.

Eyes

Eyes do not normally carry bacteria. But an eye with a mild bacterial infection may have bacteria on the eyelashes and at the corners. If employees with eye infections rub their eyes, their hands will be contaminated.

Mouth

Many bacteria are found in the mouth and on the lips. It is easy to show this by asking someone to press his or her lips on the surface of sterile agar medium in a petri dish. After 2 or 3 days at room temperature, several colonies of bacteria will grow where the lips touched the agar (see Fig. 4.1). Many bacteria and viruses found in the mouth can cause disease, especially if an employee is ill. Smoking should not be allowed at work, because the smoke can carry bacteria from the mouth, and hands pick up bacteria when they carry smoking implements to the mouth. During a sneeze, some bacteria from the mouth are released into the air. The microorganisms may infect other people or may land on food while it is being handled.

Smoking, an irritating taste in the mouth, or a head cold may make an employee want to spit. Spitting should never be allowed in a food-processing operation. Spitting spreads disease, contaminates food, and does not look good. Regular toothbrushing prevents buildup of plaque on the teeth and reduces the number of bacteria that might be carried to a food product if an employee accidentally gets saliva on his or her hands or sneezes.

Nose, Throat, and Lungs

The nose and throat have fewer bacteria than the mouth. The hairs and mucus in the upper respiratory tract filter out most of the microbes that are breathed in. The rest are destroyed by the body's defenses.

An employee who has a cold should use a hand-dip sanitizer after blowing his or her nose. Otherwise, bacteria from the nose can be transferred to the food being handled. Employees should use their elbows or shoulders to block a sneeze or cough.

A *sinus infection* means infection of the membranes in the nasal sinuses. The mucous membranes become swollen and inflamed, and secretions collect in the blocked cavities. The pressure buildup in the cavities causes pain, dizziness, and a running nose. Employees with nasal discharge should be very careful if they have to handle food products. Employees should use decongestants to reduce discharge, wash and disinfect their hands after blowing their noses, and block all sneezes with tissues.

A *sore throat* is usually caused by a type of streptococci. "Strep throat," sore throat, *laryngitis,* and *bronchitis* are spread by coughs, sneezes, and nasal discharge from carriers. Also, streptococci cause scarlet fever, rheumatic fever, and tonsillitis. These diseases spread quickly if employees practice poor hygiene.

Influenza (flu) infects the body through the lungs. Secondary bacterial infections by staphylococci, streptococci, or pneumococci can cause death.

Waste Organs

Intestinal waste (feces) is a major source of bacterial contamination. About one-third of the dry weight of stools is made up of bacteria. Particles of feces collect on the hairs around the anus and spread to underclothes. When employees use the rest room, they may get some intestinal bacteria on their hands. If they do not wash their hands properly, the microorganisms will spread to any food that they touch. Because of poor personal hygiene, the bacteria found in feces are often found in foods. Therefore, employees should wash their hands with soap before leaving the washroom, and use a hand-dip sanitizer before handling food.

The intestines of humans and animals carry common bacteria, such as *Salmonella, Shigella,* and enterococci. When these bacteria multiply, they are pathogenic (disease causing). They cause intestinal disorders and slight or severe infections, and may even cause death.

Contamination of Food Products by People

Several factors affect whether and how microbes from people contaminate food.

1. *Body site.* Different areas of the body carry different types of bacteria. The face, neck, hands, and hair are more densely populated by bacteria than are other body parts. Exposed areas of the body are more likely to be contaminated by bacteria in the air or on surfaces, or by touching other people. Hair encourages growth of microbes because of its density and oil production.
2. *Age.* The type and amount of bacteria people carry change as they grow older. These changes are most noticeable for adolescents going through puberty. Adolescents' skin glands produce large quantities of oily sebum that encourages the microorganisms that cause acne.
3. *pH.* The pH or acidity of the skin is affected by secretion of lactic acid from the sweat glands, production of fatty acids by bacteria, and diffusion of carbon dioxide through the skin. Skin usually has a pH of about 5.5. This encourages the growth of the resident microorganisms (ones that live on the skin all the time) and discourages transient microbes (ones that come and go). Soaps, creams, and other products that change the pH of the skin also alter the type of bacteria that grow.

4. *Nutrients.* Sweat contains water-soluble nutrients, and sebum contains oil-soluble nutrients. Microbiologists do not know how much these nutrients affect growth of microorganisms.

People are the biggest source of food contamination. They act as disease carriers. A carrier is a person who harbors and releases pathogens but does not have symptoms of the disease.

Handwashing

Improper handwashing causes about 25% of foodborne illnesses. Handwashing with soap and water removes bacteria. Rubbing the hands together or using a scrub brush removes more bacteria than quick handwashing.

Antimicrobial agents remove more bacteria than ordinary hand soap, but employees need to use an antimicrobial hand soap throughout the day for it to be fully effective. The antimicrobial agent needs to be in contact with the hands for more than 5 seconds to have an effect on the number of microbes. Figure 4.2 illustrates the recommended double handwashing procedure.

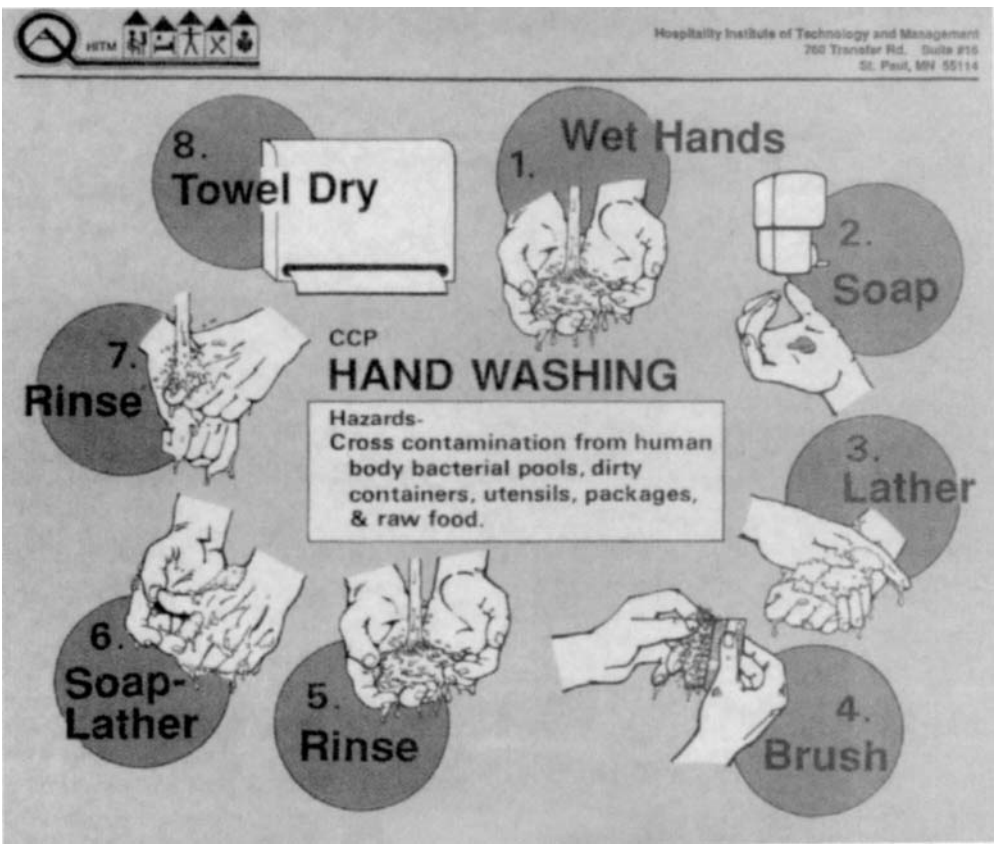


FIGURE 4.2. Recommended handwashing procedure.

Methods of Disease Transfer

Direct. Many disease microorganisms are transferred directly from one person to another through close contact. Examples are diphtheria, scarlet fever, influenza, pneumonia, smallpox, tuberculosis, typhoid fever, dysentery, and venereal diseases. Respiratory diseases are transferred via tiny particles released from the nose and mouth when a person talks, sneezes, or coughs. The particles can be suspended in the air, attached to dust, for a long time. Other people can be infected when they inhale the particles.

Indirect. The host (infected person or animal) of an infectious disease may transfer organisms to water, food, objects, or soil. Doorknobs, telephones, pencils, books, washroom fittings, clothing, money, and knives can all transfer infections from one user to another. Salmonellosis, dysentery, and diphtheria are examples of diseases that can be spread indirectly. To reduce indirect transfer of microorganisms, washbasins should have foot-operated controls instead of hand-operated faucets, and doors should be self-closing.

Requirements for Hygiene

Management must have a protocol to make sure employees use hygienic practices. Supervisors and managers should set an example for employees by using excellent hygiene and health practices themselves. They should provide proper laundry, lockerroom, and handwashing facilities to make it easy for employees to stay clean and hygienic.

All employees should have a physical examination before they are employed to check that they have good physical, mental, and emotional health. All employees who work with food should be checked regularly for signs of illness and infection, and other signs of poor health.

Employees should maintain personal hygiene in the following ways. They should:

1. Maintain good physical health through good nutrition, enough rest, and physical cleanliness
2. Report illness to their employer before working with food so that assignments can be adjusted to protect food from being contaminated
3. Practice good hygiene so that they do not contaminate food
4. Wash their hands during their work shift after using the toilet; after handling garbage or other dirty items; after handling uncooked meats, egg products, or dairy products; after handling money; after smoking; after coughing and sneezing; and when leaving or returning to food production/service areas
5. Maintain personal cleanliness through daily bathing, washing hair at least twice a week, cleaning fingernails daily, use of a hat or hairnet while handling food, and wearing clean underclothing and uniforms
6. Not touch foodservice equipment and utensils with their hands, and use disposable gloves if they have to touch food (other than dough)

7. Follow rules, such as "no smoking," and do anything else needed to protect the food from being contaminated

Employers should help employees use good hygiene in the following ways. Employers should:

1. Train employees in food handling and personal hygiene.
2. Watch and monitor employees and their work habits regularly. Employers should treat poor hygiene practices as disciplinary violations.
3. Give incentives to encourage excellent hygiene and sanitary practices.

Food handlers are responsible for their own health and personal cleanliness. Employers are responsible for making sure that the public is protected from unsanitary practices that could cause illness. Personal hygiene is essential for production of wholesome food.

HYGIENIC FOOD HANDLING

Food handlers should protect food by using barriers to separate the food from anything that could contaminate it at all stages of production. Barriers include disposable gloves, hairnets, mouth guards, sneeze guards, lids, and food wraps.

Hiring Employees

Careful hiring helps the company keep a good image and meet regulations. Over 50% of people carry *Staphylococcus aureus* in their mouths and nasal passages. The American Disease Act prohibits discrimination in hiring on the basis of disease, especially human immunodeficiency virus (HIV) status. Many carriers of disease do not have obvious symptoms. Many areas no longer require health cards because they are expensive to get, and workers can be infected after they get their card. Employees should be hired only if they meet the following conditions:

1. Applicants should not have obvious hazards, such as open sores or excessive skin infections, pimples, or acne.
2. Applicants should not have respiratory problems if they will be handling food or working in food-processing or food-preparation areas.
3. Applicants and their clothing should be neat, clean, and fresh smelling.
4. Applicants should pass a sanitation course and examination, e.g., one provided by the National Restaurant Association.

Personal Hygiene Rules

Food organizations should have clear and strict personal hygiene rules. The rules should be clearly posted on the wall or spelled out in booklets given to each employee. The policies should cover personal cleanliness, clothing, good food-handling practices, and use of tobacco.

Facilities

Employees need the right equipment and supplies to maintain good personal hygiene. Locker rooms and rest rooms should be clean, neat, well lighted, and conveniently located away from production areas. Rest rooms should have self-closing doors. Ideally, faucets at handwashing stations should be foot- or knee-operated and supply water at 43 to 50°C (109–122°F). Liquid soap dispensers are better than bars of soap, because bars can transfer microorganisms from one person to the next. Disposable towels are best for drying hands. Food and beverages should be consumed in a special area. This area should be clean and pest-free.

Supervising employees. Employees who handle food should follow the same health standards used to screen new employees. Supervisors should watch employees daily for infected cuts, boils, respiratory diseases, or other infections. Many local health departments require foodservice and food-processing firms to report employees who may have or carry an infectious disease.

Employee responsibilities. The employer is responsible for the way employees handle food, but employers should make employees responsible for the following activities when they start work. Employees should:

1. Not have respiratory, gastrointestinal, or other diseases
2. Report injuries—including cuts, burns, boils, and skin problems—to their employer
3. Report respiratory illnesses (e.g., colds, influenza, sinus infections, and bronchitis) and intestinal illnesses, such as diarrhea, to their employer
4. Bathe daily, wash their hair at least twice a week, change their underclothes daily, and keep their fingernails clean
5. Tell their supervisor when soap, towels, or other washroom supplies run out
6. Break such habits as scratching their heads or touching other parts of their body
7. Cover their mouths and noses when they cough or sneeze
8. Wash their hands after using the toilet, blowing their nose, smoking, handling anything dirty, and handling money
9. Keep their hands out of food. Food should not be tasted using a hand and should not be eaten in food production areas
10. Not use utensils that touch their mouths to handle food
11. Use disposable gloves to handle food
12. Not smoke or chew tobacco in food production and food preparation areas and should wash their hands after smoking

SUMMARY

- Microorganisms that cause illness and food spoilage often come from food handlers.
- *Hygiene* means using sanitary practices to maintain good health.

- Personal hygiene means maintaining a clean body and clean clothing.
- Parts of the body and clothing that can contaminate food include the skin, hands, fingernails, jewelry, hair, eyes, mouth, nose and lungs, and waste organs. Employees may carry a disease-causing microorganism even if they do not have symptoms of the illness.
- Microorganisms can be transferred directly (from person to person) or indirectly (via food, water, equipment, and surfaces).
- Management must hire employees who are healthy, clean, and neat. They must train employees to use good hygiene. Managers must hold employees responsible for personal hygiene so that the food that they handle is safe to eat.

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STUDY QUESTIONS

1. What should food handlers do at home to maintain good personal hygiene?
2. When should food handlers wash their hands while at work?
3. Why do healthy employees need to be careful not to contaminate food?
4. List seven body parts that are often sources of microbial infection.
5. Give two reasons why employees should not wear jewelry.
6. How can employers help employees maintain good personal hygiene?

TO FIND OUT MORE ABOUT PERSONAL HYGIENE

1. Observe a food handler (or ask someone else to watch you). Note how often they touch their face or hair, blow their nose, sneeze, and cough. Do they block their sneezes and coughs? Do they wash their hands afterwards?
2. Check the handwashing facilities at work, school, or home. Are they close to food preparation areas? Are they clean? Do they have hot and cold water, liquid soap, a nail brush, and hand towels or blow drier?

3. Put scrapings from under your fingernails, or press your lips on a sterile agar plate. Cover the plate and keep it at room temperature for 2 to 3 days. How may colonies of bacteria have grown?
4. Contact the supplier or manufacturer of the antimicrobial soap used at your workplace or school. Ask for information about good hand washing and personal hygiene.
5. Obtain a Glo-Germ kit from the Glo Germ Company, P.O. Box 537, Moab, UT 84532, (1-800) 842-6622. This kit contains an oil, a powder, and a special fluorescent lamp. The oil and powder contain plastic germs. You can put either the oil or the powder on your hands and then wash your hands normally. The fluorescent lamp shows up any remaining germs and shows how effectively you have washed your hands.



GOOD HANDWASHING *Is* IMPORTANT

These three reports show how easily poor handwashing can cause major outbreaks of foodborne illness.

On a 4-day Caribbean cruise, 72 passengers and 12 crew members had diarrhea, and 13 people had to be hospitalized. Stool samples of 19 of the passengers and 2 of the crew members contained *Shigella flexneri* bacteria. The illness was traced to German potato salad prepared by a crew member who carried these bacteria. The disease spread easily because the toilet facilities for the galley crew were limited (1).

Over 3,000 women who attended a 5-day outdoor music festival in Michigan became ill with gastroenteritis caused by *Shigella sonnei.* The illness began 2 days after the festival ended, and patients were spread all over the United States before the outbreak was recognized. An uncooked tofu salad served on the last day caused the outbreak. Over 2,000 volunteer food handlers prepared the communal meals served during the festival. Before the festival, the staff had a smaller outbreak of shigellosis. Sanitation at the festival was mostly good, but access to soap and running water for handwashing was limited. Good handwashing facilities could have prevented this explosive outbreak of foodborne illness.

Shigella sonnei caused an outbreak of foodborne illness in 240 airline passengers on 219 flights to 24 states, the District of Columbia, and 4 countries. The outbreak was identified only because it involved 21 of 65 football team players and staff. Football players and staff, airline passengers, and flight attendants with the illness all had the same strain of *S. Sonnei.* The illness was caused by cold food items served on the flights that had been prepared by hand at the airline flight kitchen. Flight kitchens should minimize hand contact when preparing cold foods or remove these foods from in-flight menus.

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(3) Hedberg, C. W., Levine, W. C., White, K. E., et al. 1992. An international foodborne outbreak of shigellosis associated with a commercial airline. *J. Am. Med Assoc.* 268: 3208.